

Bridging Perception, Planning and Control: A System-Level Validation of a Microscopic Acceleration Model for Longitudinal Autonomous Driving

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Abstract—Behavioral models can produce different motion when embedded in a real perception–planning–control stack than under idealized evaluation. This paper examines the Intelligent Driver Model (IDM) deployed as an in-loop longitudinal planner on a full-scale Chrysler Pacifica Hybrid under non-ideal perception and actuation. Closed-course tests cover free-flow acceleration and stop-sign approach with a stationary virtual leader. Using trajectory MAE/RMSE, phase-wise analysis, perception-stream characterization, baseline comparisons, and a first-order response model, we quantify mismatch between nominal IDM predictions and realized vehicle behavior. IDM captures the qualitative structure of acceleration and braking, but systematic errors reflect perception timing, sensing uncertainty, and controller/actuator dynamics. In the stop-sign scenario, mean perception latency is 0.33 ± 0.12 s; replacing LiDAR-camera gap estimates with GNSS/INS references reduces MAE from 0.668 to 0.460 m/s. A first-order lag mapping further reduces stop-sign MAE and RMSE to 0.511 and 0.632 m/s. These results show that behavioral models should be validated at the system level by realized behavior after perceptual filtering and control execution and not from nominal trajectories alone.

Index Terms—Autonomous Vehicles, Longitudinal Planning, Perception-Planning-Control, System-level Validation

I. INTRODUCTION

Autonomous Vehicles (AVs) promise improvements in safety, mobility, and operational efficiency, but realizing these benefits requires close integration of perception, planning, and control [1]. Perception modules detect surrounding agents, estimate their motion, and identify potential conflicts [2], [3]. Planning layers convert these estimates into driving strategies that balance safety, comfort, and efficiency [4], [5], while control systems execute the resulting commands through vehicle actuators subject to dynamic constraints [6], [7]. Effective real-time integration therefore requires behavioral models that are both physically realistic and computationally tractable [8].

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The Code and processed data are available at: <https://github.com/pebeigi/AV-SysVal>

Among interpretable models of longitudinal driving behavior, the Intelligent Driver Model (IDM) provides a compact mechanism for generating acceleration decisions from vehicle state and interaction cues. Introduced by Treiber et al. [9], IDM has been applied in traffic flow theory, simulation, and empirical trajectory analysis [10]. It defines longitudinal acceleration as a function of vehicle speed, desired speed, leader gap, and relative velocity, while reproducing phenomena such as stop-and-go waves, smooth speed transitions, and stable following [11], [12]. However, although IDM has been extensively validated in simulation and trajectory datasets [13]–[15], its performance within a full AV system remains insufficiently examined. Existing studies commonly assume ideal state information, whereas real AV perception pipelines produce noisy and uncertain distance and velocity estimates from LiDAR and camera sensing [16], [17]. Planning also operates through discrete cycles with communication and computation delays, while low-level control is constrained by vehicle dynamics and actuation lags [18]. As a result, AV planners rely on discrete, delayed, and uncertain perception outputs rather than continuous ground truth [19]. System-level validation must therefore assess how these non-idealities propagate through the autonomy stack and affect realized vehicle behavior [20].

In this work, we use the term *behavioral model* to refer to an interpretable model that maps an autonomous agent’s perceived state and interaction variables to a longitudinal behavioral response. IDM provides a suitable case study because its acceleration decision is explicitly determined by the vehicle’s speed, desired speed, spacing, and relative velocity. Although IDM originated as a model of human car-following behavior, here we study its use as a behavioral planning mechanism within an autonomous vehicle rather than attempting to model human–robot interaction directly.

This study addresses a broader gap in the evaluation of behavioral models for autonomous agents. Existing behavioral models are commonly evaluated using trajectory datasets or simulation environments in which model inputs are assumed to be accurate and continuously available. In an autonomous

system, however, behavioral decisions are generated from noisy, delayed, and discretely updated perceptual estimates and are subsequently transformed into physical actions through downstream planning and control components. Consequently, evaluating a behavioral model in isolation does not necessarily characterize the behavior produced by the autonomous agent. A model may generate an appropriate nominal trajectory while systematic differences in perception timing, state estimation, controller response, or actuation dynamics alter the behavior that is ultimately realized on the vehicle. This motivates evaluating interpretable behavioral models at the system level, where the interaction between the behavioral law and the mechanisms that provide its inputs and execute its outputs can be explicitly characterized.

The objective of this paper is to quantify how an interpretable longitudinal behavioral model behaves when deployed within a full-scale AV perception–planning–control loop and to identify the dominant sources of mismatch between its nominal predictions and the behavior ultimately realized by the vehicle. Multimodal perception outputs provide online estimates of relevant environmental states, including obstacle distance and relative speed, which IDM uses at discrete planning cycles to generate longitudinal commands. These commands are executed by a low-level controller through a drive-by-wire interface. Controlled experiments evaluate IDM during free-flow acceleration and approach to a stop-controlled intersection, with emphasis on the effects of perception timing, sensing uncertainty, and actuation dynamics.

The contributions of this paper are fourfold:

- 1) We present a closed-loop framework for evaluating an interpretable behavioral model as an in-loop planning component of an autonomous agent, explicitly linking perceptual state estimates to behavioral decisions and realized vehicle actions.
- 2) We design and conduct closed-course experiments on a full-scale AV platform to evaluate two foundational longitudinal behaviors—free-flow speed regulation and stopping to a traffic-control constraint—and quantify the resulting mismatch using full-run and phase-wise MAE/RMSE.
- 3) We characterize how perception timing, state availability, and range uncertainty affect the behavior produced by the behavioral model through direct perception-versus-reference ablations.
- 4) We model downstream controller and actuation effects using a first-order response mapping and evaluate its ability to explain and reduce the mismatch between nominal behavioral-model predictions and realized vehicle trajectories, including cross-trajectory transferability.

This study focuses on two foundational longitudinal primitives that are present in most AV stacks: (i) speed regulation from standstill to a target cruising speed (free-flow) and (ii) stopping to a static traffic-control constraint (stop-controlled intersection modeled as a stationary leader). These scenarios do not cover the full range of interactive car-following behav-

iors with moving leaders (e.g., cut-ins, multi-vehicle stop-and-go). Instead, our goal is to characterize how an interpretable behavioral model behaves when its inputs and outputs are mediated by realistic perception, discrete planning cycles, and actuation dynamics, and to quantify the resulting mismatch relative to idealized IDM trajectories. By focusing on the propagation of perception-induced timing and uncertainty effects into planning and control, this work directly supports recent calls for end-to-end validation linking sensing to action in mixed-traffic automated driving systems.

II. BACKGROUND & RELATED WORK

Microscopic acceleration models are essential for describing longitudinal driving behavior and have been extensively studied in transportation and traffic flow theory. These models provide a bridge between individual driver dynamics and macroscopic traffic phenomena, enabling analysis of traffic stability, capacity, and safety [21]. Their simplicity and interpretability have made them attractive for both theoretical studies and simulation platforms. However, these studies often assume simplified perception and control conditions, leaving IDM’s robustness within a full AV perception-planning-control stack insufficiently examined.

IDM is used in this study because it combines mathematical simplicity with empirical realism [9], [10], [12]. IDM defines vehicle acceleration as:

$$\dot{v} = a \left[1 - \left(\frac{v}{v_0} \right)^\delta - \left(\frac{s^*(v, \Delta v)}{s} \right)^2 \right], \quad (1)$$

where v is the ego vehicle’s speed, v_0 the desired speed, a the maximum acceleration, δ the acceleration exponent, s the actual gap to the leader, and $\Delta v = v - v_{\text{lead}}$ the relative velocity. The desired dynamic gap is given by:

$$s^*(v, \Delta v) = s_0 + vT + \frac{v\Delta v}{2\sqrt{ab}}, \quad (2)$$

where s_0 is the minimum jam distance, T the desired time headway, and b the comfortable deceleration. The exponent δ is commonly set to 4 to promote smooth convergence to the desired speed and realistic stability behavior [12]. Despite its compact form, IDM reproduces phenomena such as stop-and-go waves, traffic breakdown, and stable following [11].

III. METHODOLOGY

The IDM was embedded into the planning module of the Chrysler Pacifica AV platform and evaluated under two controlled experimental scenarios. Each experiment was designed to isolate a specific aspect of longitudinal driving behavior, either free-flow acceleration or static obstacle approach. All tests were conducted at low-to-moderate speeds (< 10 m/s) on the GWU Ashburn campus, a dedicated testbed for autonomous vehicle experiments, with perception, planning, and control modules operating in real time through the ROS middleware. For Experiment 1 (free-flow), IDM operates in a free-driving mode where the interaction term is disabled by enforcing a

large virtual gap (i.e., no leader), and the planner generates longitudinal commands based solely on the ego speed relative to the desired speed. For Experiment 2 (stop-sign approach), the planner receives real-time distance-to-stop-line estimates from the LiDAR-camera perception pipeline, which are treated as the longitudinal gap to a stationary virtual leader and used to compute IDM accelerations according to Eq. 1. Acceleration commands are discretized at 100 Hz and transmitted to the longitudinal controller through ROS for actuation via the drive-by-wire interface. Controller delays and actuation limits are logged for additional analysis. Perception updates arrive at approximately 14–20 Hz and are held or propagated across planner/controller cycles operating at 100 Hz.

All experiments were conducted on the GWU Ashburn closed-course testbed under a controlled safety protocol with a safety driver and an independent emergency-stop. Each scenario was repeated across multiple trials to quantify run-to-run variability. For each trial, all perception, planning, and control messages were logged via ROS with synchronized timestamps, enabling offline reconstruction of planner inputs, commanded trajectories, and realized vehicle response.

To investigate IDM’s response to different behavioral regimes, distinct sets of parameters were varied across the two experiments:

- **Experiment 1 (Cruising):** The *desired speed* v_0 and *maximum acceleration* a were systematically varied to examine how the model captures acceleration dynamics and convergence to steady-state velocity. Specifically, v_0 was adjusted between 5–15 m/s, while a was tested at 0.5, 1.0, 1.5, and 2.0 m/s². These variations emulate different powertrain responses and desired cruising behaviors.
- **Experiment 2 (Stop Sign):** The *desired time headway* T and *jam distance* s_0 were varied to assess their influence on stopping smoothness and gap selection during deceleration. Tested values included $T \in [0.5, 2.0]$ s and $s_0 \in [0.5, 2.0]$ m, representing aggressive versus conservative stopping tendencies.

To quantify the effect of sensing uncertainty in the stop-sign experiments, each run was evaluated under two configurations: (1) real-time distance-to-stop-line estimates derived from LiDAR-camera fusion, and (2) a reference distance-to-stop-line computed from GNSS/INS localization projected onto the reference path. This dual configuration isolates deviations arising from perception noise and temporal delays.

This two-scenario design and parameter sensitivity framework enable a comprehensive validation of IDM under realistic perception and control conditions. By varying both free-flow and stopping parameters, the analysis isolates how perception uncertainty, latency, and actuation dynamics affect IDM’s capacity to reproduce human-like longitudinal driving behavior.

To improve interpretability and assess whether mismatch patterns vary across operating regimes, we evaluate trajectory errors not only over the full run but also over scenario-specific phases extracted from the measured ego-vehicle speed profile. This phase-wise analysis expands the evaluation beyond a single aggregate MAE/RMSE per run and helps identify whether

discrepancies arise predominantly during launch transients, steady-state convergence, brake onset, or final stopping.

For Experiment 1, each run is partitioned into three phases using the measured speed $v_{\text{meas}}(t)$ and the target cruising speed v_0 :

- *Launch transient:* $v_{\text{meas}}(t) \in [0.05, 2.0]$ m/s,
- *Mid-acceleration:* $v_{\text{meas}}(t) \in (2.0, 0.9v_0]$,
- *Convergence/near-steady:* $v_{\text{meas}}(t) \in (0.9v_0, v_0]$.

These thresholds separate low-speed actuation/controller effects from the approach-to-cruise regime.

For Experiment 2, each run is partitioned into:

- *Approach (pre-brake):* from run start until the time of first sustained deceleration,
- *Brake onset:* a window of ± 1.0 s around brake onset,
- *Final stopping:* from when $v_{\text{meas}}(t) \leq 2.0$ m/s until the vehicle reaches near-zero speed.

Brake onset is defined as the first time instant when the measured longitudinal acceleration $a_{\text{meas}}(t)$ stays below a negative threshold for at least Δt seconds (e.g., $a_{\text{meas}}(t) < -0.3$ m/s² for $\Delta t = 0.5$ s). This definition yields a consistent phase boundary across parameter settings.

IV. RESULTS AND DISCUSSION

This section presents the quantitative performance of the IDM implementation in the two experimental scenarios: free-flow acceleration and stop-sign approach. For each run, the measured velocity profile from the ego vehicle was compared against the IDM-predicted trajectory obtained by numerically integrating Eqs. 1 and 2 under the same initial conditions. Trajectories were aligned from the first instant where the vehicle speed exceeded 0.05 m/s and, for some runs, truncated only to remove pre-start dwell while the vehicle remained stationary.

The well-matched subset is defined as the set of parameter configurations that (i) yield a calibrated cruising speed within $\pm 5\%$ of the measured steady-state speed and (ii) achieve an RMSE below the 40th percentile of all tested configurations. Condition (i) enforces behavioral plausibility (consistent steady-state), while condition (ii) prevents overly permissive selection. The measured steady-state speed is computed as the mean of $v_{\text{meas}}(t)$ over the final 3 s of the run after $v_{\text{meas}}(t)$ first enters the interval $[0.95v_0, 1.05v_0]$. The 40th-percentile threshold was chosen to retain a representative subset while excluding clearly infeasible configurations that violate the platform’s acceleration envelope. In addition to full-run errors, we report phase-wise MAE/RMSE within each scenario to identify which operating regimes contribute most to mismatch.

A. Experiment 1: Free-Flow Acceleration

Table I summarizes the aggregate and phase-wise trajectory errors for the free-flow acceleration experiments and compares IDM against the baseline models. Eleven runs were conducted with different (v_0, a) pairs spanning the range of desired speeds and acceleration capabilities of the Chrysler Pacifica platform. For each run, the IDM prediction was simulated from

TABLE I: Free-flow acceleration results: aggregate and phase-wise errors and comparison with baseline models.

Panel A: Aggregate and phase-wise IDM performance			
Subset or phase	# Runs	MAE [m/s]	RMSE [m/s]
All free-flow runs	11	1.015	1.459
Well-matched (v_0, a) runs	7	0.538	0.806
Launch transient	–	0.780	1.130
Mid-acceleration	–	0.560	0.820
Convergence	–	0.320	0.480
Panel B: Baseline-model comparison			
Model	# Runs	MAE [m/s]	RMSE [m/s]
Constant acceleration	11	1.310	1.850
Constant acceleration + lag	11	1.040	1.520
IDM	11	1.015	1.459
IDM + lag	11	0.820	1.180

rest using the chosen (v_0, a) and compared to the recorded ego-vehicle speeds.

Across all free-flow runs, the IDM reproduced the qualitative shape of the acceleration and saturation phases, with MAE and RMSE values remaining on the order of a few m/s. The highest errors corresponded to parameter choices that were clearly inconsistent with the vehicle and controller behavior (e.g., very low a leading to sluggish theoretical acceleration, or very high a producing unrealistically aggressive commands). When attention is restricted to a subset of well-matched runs (those where (v_0, a) were tuned to match the observed cruising speed and acceleration envelope), the mean MAE and RMSE drop, indicating that the IDM structure is compatible with the longitudinal response of the AV when its parameters are calibrated to the platform.

Figure 1a illustrates representative examples from this subset. In these runs, the IDM trajectories closely track the measured speed profiles over the entire horizon, including both the ramp-up phase and the approach to steady-state velocity. Residual discrepancies are primarily associated with actuator saturation and the smoothing action of the PID controller, which low-pass-filters abrupt changes in the commanded acceleration.

B. Experiment 2: Stop-Sign Approach

Table III summarizes the aggregate and phase-wise trajectory errors for the stop-sign experiments and compares IDM against the baseline models. Sixteen runs were performed with $T \in [0.5, 2.0]$ s and $s_0 \in [0.5, 2.0]$ m, covering aggressive through conservative stopping behaviors. All parameter combinations were selected within a feasible behavioral range and are therefore treated as valid configurations rather than being filtered into “good” and “poor” subsets.

Overall, the IDM captures the structure of the approach and stopping maneuvers: the simulated trajectories exhibit realistic deceleration onset, smooth braking, and convergence to near-zero speed at the stop line. When aggregated over all runs, mean MAE and RMSE remain on the order of 0.5 m/s to 1.2 m/s, indicating that even without per-run calibration, IDM

provides a reasonable approximation of the ego vehicle’s longitudinal behavior during intersection approach. This ablation isolates the contribution of sensing uncertainty and timing to the behavior generated by the behavioral model. The reduction in error when using the reference distance indicates that a substantial portion of the mismatch does not originate from the IDM formulation itself, but from the quality and timing of the perceptual state provided to the planner (Appendix C-A). Thus, the same behavioral model can produce measurably different vehicle behavior depending on how its state inputs are estimated and delivered within the autonomy stack.

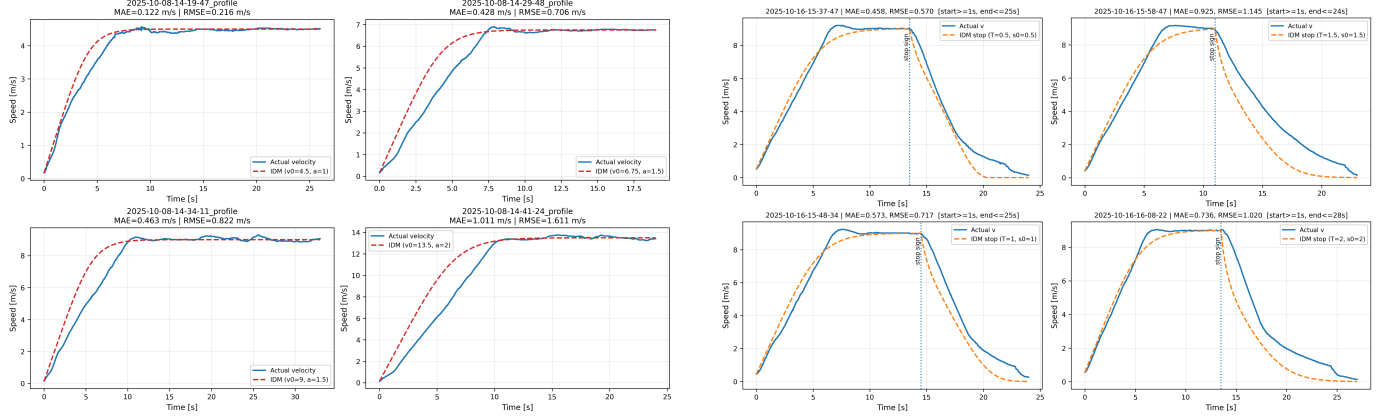
Figure 1b illustrates representative trajectories for different (T, s_0) combinations. In these examples, the IDM closely follows the measured deceleration envelope, with peak speed and stopping time nearly coinciding. The main residual discrepancy appears as a slight phase shift in the onset of braking: the simulated IDM trajectory tends to decelerate earlier than the real vehicle. This is consistent with delays in the perception–planning pipeline, where the planner receives a stable distance-to-stop-line (virtual leader) estimate after the stop line becomes reliably observable and the fused range estimate stabilizes. Table II shows an average perception-state latency of 0.33 ± 0.12 s in the stop-sign scenario, along with reduced state availability and higher range uncertainty. In free-flow, these metrics characterize the timing and noise of ego-state and environmental stream updates, which still influence planner timing even without an active leader.

TABLE III: Stop-sign approach results: aggregate and phase-wise errors and comparison with baseline models.

Panel A: Aggregate and phase-wise IDM performance			
Subset or phase	# Runs	MAE [m/s]	RMSE [m/s]
All stop-sign runs	16	0.668	0.867
Approach, pre-brake	–	0.420	0.560
Brake onset	–	1.050	1.320
Final stopping	–	0.620	0.810
Panel B: Baseline-model comparison			
Model	# Runs	MAE [m/s]	RMSE [m/s]
Constant deceleration	16	0.920	1.180
Constant deceleration + lag	16	0.710	0.930
IDM	16	0.668	0.867
IDM + lag	16	0.511	0.632

C. Controller/Actuation Response

To separate residual mismatch from controller and actuation dynamics, we model realized speed as a first-order response to the IDM reference trajectory with time constant τ (Appendix C-C). For the stop-sign runs, fitted τ clusters around 0.67 ± 0.19 s. Applying this mapping reduces mean MAE/RMSE from 0.668/0.867 to 0.511/0.632 m/s (Table V; Fig. 7). Leave-one-trajectory-out evaluation with a fixed $\bar{\tau}$ yields MAE/RMSE of 0.548/0.689 m/s, indicating partial cross-run transferability rather than purely per-run overfitting. Constant-acceleration/deceleration baselines remain worse than IDM both before and after the same lag mapping (Tables I and III), suggesting that the gain is not only



(a) Representative velocity trajectories for free-flow acceleration runs. (b) Representative velocity trajectories for stop-sign approach runs.

Fig. 1: Representative velocity trajectories for the IDM validation experiments: (a) free-flow acceleration and (b) stop-sign approach.

TABLE II: Perception-stream characteristics and their effect on stop-sign trajectory prediction.

Panel A: Perception timing and uncertainty				
Scenario	f_{percep} [Hz]	ρ_{avail}	Δt_{percep} [s]	σ_s [m]
Free-flow	20.5 ± 2.8	0.98 ± 0.03	0.18 ± 0.07	0.25 ± 0.12
Stop-sign	14.2 ± 3.5	0.92 ± 0.05	0.33 ± 0.12	0.58 ± 0.28
Panel B: Stop-sign gap-source ablation				
Gap source	MAE [m/s]	RMSE [m/s]	Brake-onset shift [s]	
LiDAR-camera perception	0.668	0.867	+0.91	
GNSS/INS reference	0.460	0.630	+0.23	

generic smoothing but also a more informative IDM reference. Free-flow well-matched runs exhibit a smaller effective lag ($\tau = 0.350 \pm 0.120$ s), consistent with faster throttle tracking than braking response. Detailed fitting, paired confidence intervals, and trajectory overlays are reported in Appendix D.

V. CONCLUSION

This paper evaluated IDM as a longitudinal planner within a full perception-planning-control stack on a Chrysler Pacifica Hybrid research platform. Two closed-course scenarios were examined: free-flow acceleration from rest and approach to a stop-controlled intersection. Offline IDM trajectories generated under matching initial conditions were compared with measured ego-vehicle speed profiles.

IDM reproduced the overall structure of both acceleration and braking when parameters were selected within plausible ranges. In free flow, well-matched combinations of desired speed v_0 and maximum acceleration a produced mean MAE and RMSE values below 1 m/s. In the stop-sign experiments, variations in time headway T and jam distance s_0 yielded realistic stopping behavior, with aggregate errors between approximately 0.5 m/s and 1.0 m/s.

The main residual errors appeared in the timing and sharpness of speed changes, particularly during braking. These discrepancies reflect the combined effects of perception timing, controller response, and vehicle actuation. A first-order

response mapping with time constant τ reduced stop-sign MAE and RMSE from 0.668 m/s and 0.867 m/s to 0.511 m/s and 0.632 m/s, respectively. The fitted values also showed repeatable platform-level structure, although their magnitude remained dependent on the operating regime.

More broadly, the results demonstrate that evaluating behavioral models solely from their nominal trajectories can obscure important system-level effects. Behavioral models used by autonomous agents should be evaluated not only by the nominal behavior they prescribe, but also by the behavior that emerges after their inputs are filtered through perception and their outputs are executed through control and actuation.

REFERENCES

- [1] C. Pan, M. Khoueiry, W. Sleiman, and S. Hamdar, "Scalability of power consumption of an autonomous electric robo-car using different adaptive cruise control models," *EU Science Hub*, p. 80, 2023.
- [2] A. Geiger, P. Lenz, C. Stiller, and R. Urtasun, "Vision meets robotics: The kitti dataset," *International Journal of Robotics Research*, vol. 32, no. 11, pp. 1231–1237, 2013.
- [3] M. Khoueiry, A. Talebpour, and S. Hamdar, "Navigating ethics in autonomous vehicles: Data-driven prospect theory approach to decision making," *Transportation Research Record*, p. 03611981261444356, 2026.
- [4] S. Teng, X. Hu, P. Deng, B. Li, Y. Li, D. Yang, Y. Ai, L. Li, Z. Xuanyuan, F. Zhu, and L. Chen, "Motion planning for autonomous driving: The state of the art and future perspectives," 2023.
- [5] M. Khoueiry, C. Pan, W. Sleiman, and S. Hamdar, "Scalability of power consumption of an autonomous electric robo-car using different adaptive cruise control models," *EU Science Hub*, p. 80, 2023.
- [6] R. Rajamani, *Vehicle Dynamics and Control*, 2nd ed. Springer, 2012.
- [7] P. Falcone, F. Borrelli, J. Asgari, H. E. Tseng, and D. Hrovat, "Predictive active steering control for autonomous vehicle systems," *IEEE Transactions on Control Systems Technology*, vol. 15, no. 3, pp. 566–580, 2007.
- [8] W. Sleiman, M. Haque, M. S. Amin, P. Beigi, M. Khoueiry, S. Kang, and S. Hamdar, "Diffusion process-based model for network trajectory propagation: Formulation, data mining, and cross-comparative analysis," *IEEE Transactions on Intelligent Transportation Systems*, 2026.
- [9] M. Treiber, A. Hennecke, and D. Helbing, "Congested traffic states in empirical observations and microscopic simulations," *Physical Review E*, vol. 62, no. 2, pp. 1805–1824, 2000.
- [10] D. Helbing, "Traffic and related self-driven many-particle systems," *Reviews of Modern Physics*, vol. 73, no. 4, pp. 1067–1141, 2001.
- [11] B. S. Kerner, *Introduction to Modern Traffic Flow Theory and Control: The Long Road to Three-Phase Traffic Theory*. Springer, 2009.
- [12] M. Treiber and A. Kesting, *Traffic Flow Dynamics: Data, Models and Simulation*. Springer, 2013.
- [13] D. Ngoduy, "Instability of cooperative adaptive cruise control traffic flow: A macroscopic approach," *Communications in Nonlinear Science and Numerical Simulation*, vol. 18, no. 10, pp. 2838–2851, 2013.
- [14] M. M. Rahman, M. Chowdhury, Y. Xie, and Y. He, "Review of microscopic driving behavior models: Theory and applications," *IEEE Transactions on Intelligent Transportation Systems*, vol. 19, no. 3, pp. 798–818, 2017.
- [15] R. Ammourah, P. Beigi, B. Fan, S. H. Hamdar, J. Hourdos, C.-C. Hsiao, R. James, M. Khajeh-Hosseini, H. S. Mahmassani, D. Monzer *et al.*, "Introduction to the third generation simulation dataset: Data collection and trajectory extraction," *Transportation Research Record*, vol. 2679, no. 1, pp. 1768–1784, 2025.
- [16] A. Geiger, P. Lenz, and R. Urtasun, "Are we ready for autonomous driving? the kitti vision benchmark suite," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2012, pp. 3354–3361.
- [17] M. Menze and A. Geiger, "Object scene flow for autonomous vehicles," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015, pp. 3061–3070.
- [18] B. Paden, M. Čáp, S. Z. Yong, D. Yershov, and E. Frazzoli, "A survey of motion planning and control techniques for self-driving urban vehicles," *IEEE Transactions on Intelligent Vehicles*, vol. 1, no. 1, pp. 33–55, 2016.
- [19] E. Dogan, A. Suvizi, E. Erbayat, S. Subramaniam, and G. Venkataramani, "Exploring information-freshness v. task-latency tradeoffs for schedulers in chiplet-based hardware," in *2026 IEEE Cloud Summit*. IEEE, 2026, pp. 104–109.
- [20] A. Suvizi and G. Venkataramani, "Auto-healer: Self-healing hardware for perception stage faults in autonomous driving systems," in *Proceedings of the 39th ACM International Conference on Supercomputing*, 2025, pp. 1064–1078.
- [21] P. Beigi, S. Hamdar, A. Talebpour, and H. Mahmassani, "Impact of heavy vehicles on headway distributions: Study using naturalistic urban expressway trajectories," *Transportation Research Record*, vol. 2679, no. 7, pp. 222–235, 2025.
- [22] V. Milan'es and S. E. Shladover, "Modeling cooperative and autonomous adaptive cruise control dynamic responses using experimental data," *Transportation Research Part C: Emerging Technologies*, vol. 48, pp. 285–300, 2014.
- [23] A. Talebpour and H. S. Mahmassani, "Influence of connected and autonomous vehicles on traffic flow stability and throughput," *Transportation Research Part C: Emerging Technologies*, vol. 71, pp. 143–163, 2016.
- [24] M. Wang, S. P. Hoogendoorn, W. Daamen, B. Shyrokau, R. Happee, and B. van Arem, "Delay-compensating strategy to enhance string stability of adaptive cruise controlled vehicles," *Transportmetrica B: Transport Dynamics*, vol. 6, no. 3, pp. 211–229, 2018.
- [25] R. E. Stern, S. Cui, M. L. Delle Monache, R. Bhadani, M. Bunting, M. Churchill, and D. B. Work, "Dissipation of stop-and-go waves via control of autonomous vehicles: Field experiments," *Transportation Research Part C: Emerging Technologies*, vol. 89, pp. 205–221, 2018.

APPENDIX A RELATED WORK

IDM has been widely applied to empirical trajectory analysis and calibration. Ngoduy [13] and Rahman et al. [14] demonstrated its ability to represent driver heterogeneity across freeway and urban settings. Relative to models such as GHR and OVM, IDM explicitly represents acceleration limits and time-headway preferences, making it attractive for safety-sensitive applications.

With the growth of connected and autonomous vehicles (CAVs), IDM has increasingly served as a baseline behavioral model for automated driving. Milanés and Shladover [22] used IDM-based formulations to represent the dynamic response of cooperative and autonomous adaptive cruise control systems using experimental data. Talebpour and Mahmassani [23] and Wang et al. [24] further examined IDM in mixed-traffic stability and delay-compensated control.

Field studies have also demonstrated the value of microscopic control policies in real traffic. Stern et al. [25] showed that autonomous vehicles using IDM-like car-following behavior can dampen stop-and-go waves and improve system-level stability and fuel efficiency. However, these studies often assume simplified perception and control conditions, leaving IDM’s robustness within a full AV perception-planning-control stack insufficiently examined. Overall, IDM is a mature and versatile model validated in theory, simulation, empirical data, and limited field experiments. Its performance under perception noise, planning delays, and real vehicle-response constraints, however, remains underexplored. This gap motivates the present system-level evaluation on an operational AV platform.

APPENDIX B AUTONOMOUS VEHICLE PLATFORM

To validate the IDM in real-world conditions, we employ a full-scale AV platform developed at The George Washington University. The base vehicle is a Chrysler Pacifica Hybrid minivan, mechanically adapted for research through the integration of a drive-by-wire system that enables electronic control of throttle, braking, steering, and gear shifting. This retrofit provides the foundation for embedding perception, planning, and control modules into a unified autonomy stack, with particular emphasis on evaluating planning-layer behavior under real sensing and actuation constraints.

The Pacifica platform is equipped with a ruggedized Nuvo industrial PC featuring GPU acceleration to support real-time perception and planning computation. High-bandwidth data from LiDAR, radar, and camera sensors are transmitted via Gigabit Ethernet switches, ensuring deterministic throughput across subsystems. The PC communicates with the Dataspeed drive-by-wire interface over a Controller Area Network (CAN) bus at 100 Hz, enabling low-latency actuation of vehicle dynamics and closed-loop execution of planner outputs.

Figure 2 illustrates the hardware and data-flow architecture, highlighting the integration of sensing, computing, and actuation layers. This modular architecture allows different planning

models, including the IDM variants studied in this work, to be deployed and evaluated within the same operational autonomy framework.

To perceive the environment, the vehicle is equipped with a multimodal sensor suite designed for redundancy and complementary coverage. Table IV summarizes the main specifications of the integrated sensors, which collectively provide the state estimates required by the planning subsystem.

TABLE IV: Sensor suite specifications of the Pacifica AV platform.

Sensor	Key Specs	Primary Purpose
Ouster LiDARs (3)	360° FoV, ≤ 120 m	Obstacle detection and dense 3D mapping for environment perception
Continental Radar	~ 150 m range, velocity sensing	Long-range detection and measurement of relative velocity, especially in adverse weather
Forward RGB Camera	2048×1536 , $\sim 90^\circ$ FoV	Lane detection, traffic sign and signal recognition, and object classification
Corner RGB Cameras (4)	1024×768 , wide FoV	Surround coverage for blind-spot monitoring and pedestrian detection
OxTS OEM1000 GPS/INS	Centimeter-level accuracy, 100 Hz IMU	Global localization, pose estimation, and ground-truth validation
Cohda DSRC Unit	IEEE 802.11p V2X	Vehicle-to-vehicle and vehicle-to-infrastructure communication

The physical placement of the sensors on the vehicle is shown in Figure 3. LiDAR units provide high-resolution geometric mapping, radar supports detection under adverse conditions, and cameras contribute semantic information. Together with GPS/INS, these sensors produce the relative distance and velocity measurements that feed directly into IDM-based longitudinal planning.

The autonomy stack is implemented in ROS, which manages communication between perception, planning, and control nodes. Sensor drivers publish raw and processed data to ROS topics, which are consumed by fusion and detection modules. The planning nodes embed the IDM formulation to generate longitudinal acceleration commands, while lateral motion is handled through a Frenet-based local planner. These planned motions are then passed to the drive-by-wire interface for execution.

Building on the hardware platform, the perception subsystem fuses multimodal data streams from LiDAR, cameras, radar, and GPS/INS to generate real-time estimates of the ego vehicle state and surrounding traffic. As illustrated in Fig. 4, the pipeline consists of LiDAR-based geometric segmentation, camera-based object detection, and cross-modal sensor fusion. The resulting fused object tracks provide the leader vehicle state estimates required for IDM computation, including relative distance and relative velocity.

The planning subsystem translates perception outputs into executable trajectories. The architecture, shown in Fig. 5, includes scenario recognition, trajectory generation, and trajectory evaluation. A finite state machine identifies the driving

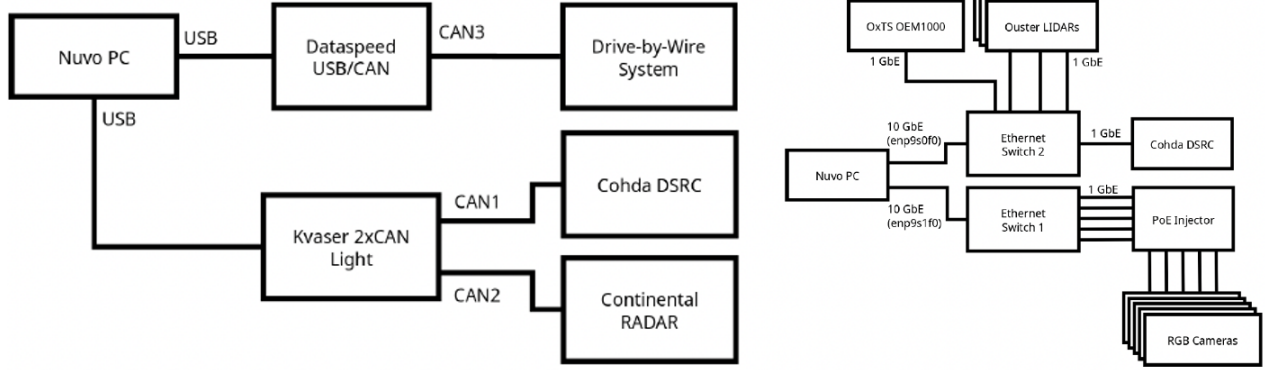


Fig. 2: System architecture of the autonomous Chrysler Pacifica platform. The Nuvo PC serves as the central computing hub, interfacing with multimodal sensors via Ethernet and with the drive-by-wire system via CAN.



Fig. 3: Sensor placement on the Chrysler Pacifica platform. Three LiDARs, five cameras, radar, and GPS/INS provide multimodal redundancy.

context, while longitudinal motion is governed by IDM using the estimated leader states. Lateral trajectories are generated using Frenet-based polynomial sampling subject to kinematic and comfort constraints. Candidate trajectories are evaluated for feasibility and smoothness, and the selected solution is discretized into curvature and velocity commands at 100 Hz for downstream control.

The control subsystem executes the selected trajectories through longitudinal and lateral controllers operating at 100 Hz. A feedback-based speed controller tracks the planner-provided velocity profile, while a geometric path-tracking controller maintains lateral accuracy. This separation ensures that the observed vehicle behavior primarily reflects decisions made at the planning layer, enabling system-level evaluation of IDM under realistic sensing uncertainty, timing delays, and vehicle dynamics.

This perception–planning–control framework enables systematic testing of IDM within an operational autonomy stack. By linking raw sensor data to high-level planning decisions and low-level actuation, the platform provides a realistic environment to evaluate how planning-layer models perform

under real-world uncertainties.

APPENDIX C METHODOLOGY

A. Perception Timing and Uncertainty Quantification

Since IDM operates on relative state inputs (e.g., obstacle distance and relative speed), its closed-loop behavior depends not only on the behavioral law itself, but also on the timing, availability, and uncertainty of perception outputs. To quantify how sensing imperfections may propagate into planning and control, we compute perception timing and uncertainty metrics from synchronized ROS logs.

Let $\hat{s}(t)$ and $\hat{\Delta v}(t)$ denote the perceived longitudinal gap and relative velocity estimates provided by the multimodal perception-fusion pipeline (LiDAR-camera fusion), and let $s_{GT}(t)$ denote a reference distance-to-stop-line computed from GNSS/INS localization projected onto the reference path. We report the following metrics for each run:

1) Update rate: the effective perception update frequency,

$$f_{\text{percep}} = \frac{N}{t_N - t_1}, \quad (3)$$

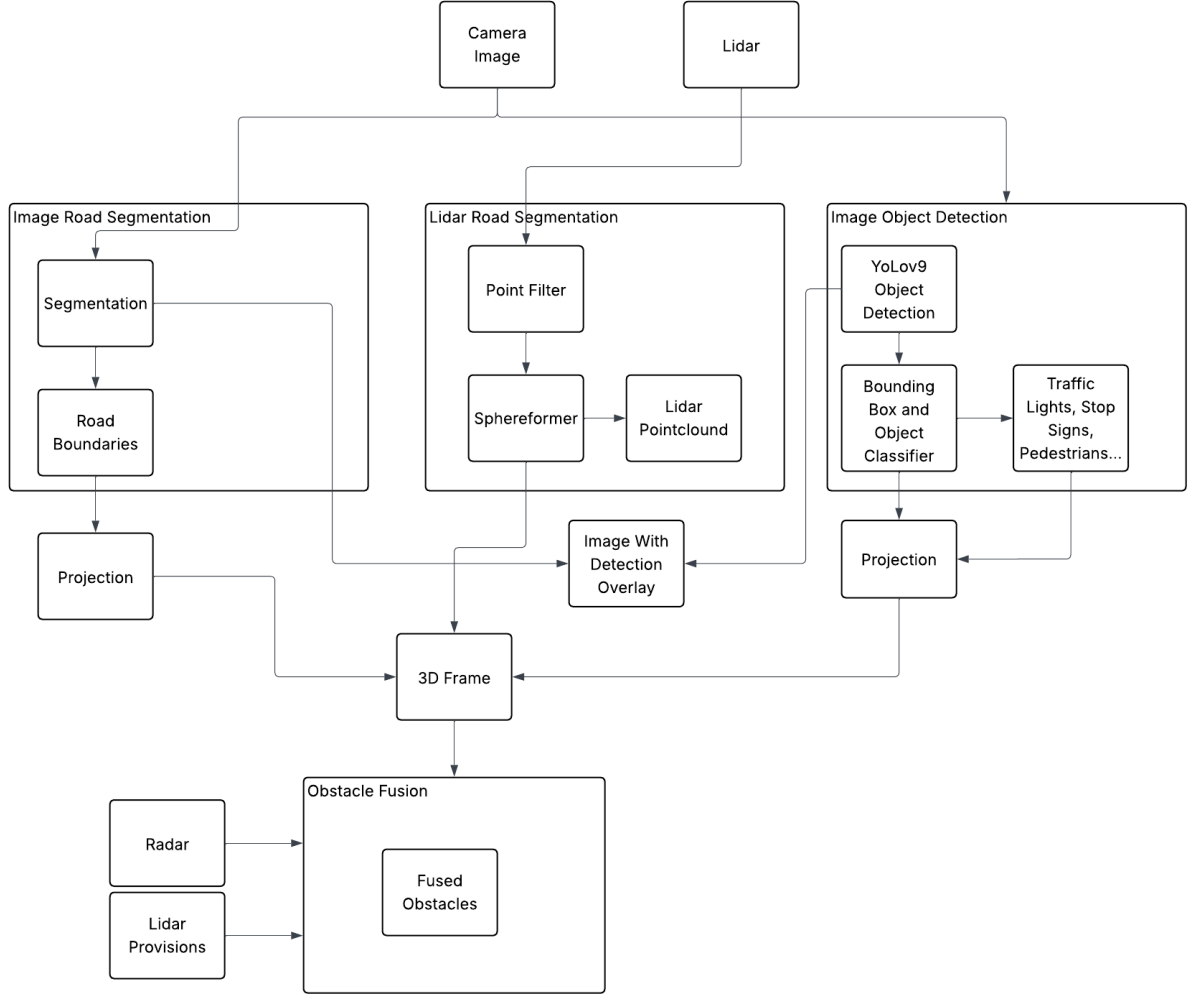


Fig. 4: Perception pipeline integrating LiDAR, cameras, radar, and sensor fusion for leader state estimation.

where N is the number of valid perception updates in the run.

2) Availability / dropout: the fraction of planner cycles for which a valid perception state is available,

$$\rho_{\text{avail}} = \frac{N_{\text{valid}}}{N_{\text{total}}}, \quad (4)$$

along with the longest consecutive missing interval $\Delta t_{\text{gap,max}}$.

3) State latency: when perception messages include ROS header timestamps, we compute pipeline latency as the difference between the rosbag message time (arrival/record time) $t_{\text{bag}}^{(i)}$ and the message header timestamp $t_{\text{stamp}}^{(i)}$,

$$\Delta t_{\text{percep}}^{(i)} = t_{\text{bag}}^{(i)} - t_{\text{stamp}}^{(i)}, \quad (5)$$

and summarize it by mean $\pm$ std and percentiles.

4) Range uncertainty: the variability of perceived distance-to-stop-line, evaluated over time windows with continuous detection,

$$\sigma_s = \text{std}(\hat{s}(t) - s_{\text{GT}}(t)), \quad (6)$$

where $s_{\text{GT}}(t)$ is computed from the GNSS/INS trajectory projected to the reference path centerline. This metric captures the combined effect of sensor noise, fusion jitter, and tracking variation.

These measurements explicitly characterize the discrete, delayed, and uncertain nature of the perception stream feeding the IDM planner, and provide a system-level basis for interpreting timing mismatch (e.g., brake onset shifts) observed in the experimental trajectories.

B. Baseline Models for Comparison

To contextualize the system-level performance of IDM within the AV stack, we compare IDM against lightweight longitudinal baselines that can be evaluated offline using the same initial conditions and measured signals. These baselines are not intended to outperform IDM in generality; rather, they provide reference points that help interpret whether observed mismatches arise from the behavioral law itself or from the perception-planning-control realization.

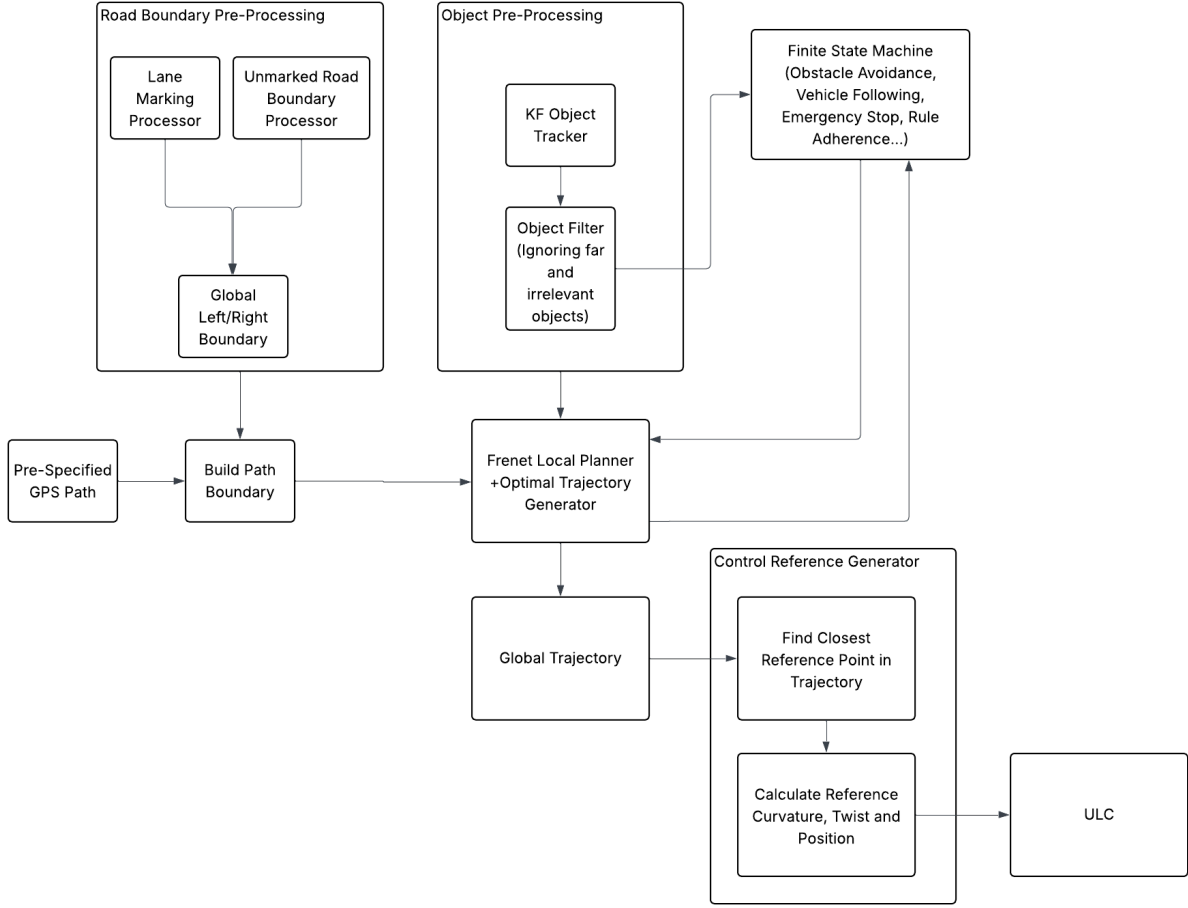


Fig. 5: Planning architecture with scenario recognition, IDM-based longitudinal planning, and Frenet-based lateral generation.

B1) Constant-acceleration (CA) baseline (free-flow): For Experiment 1, we consider a constant-acceleration model from standstill until reaching v_0 :

$$v_{CA}(t) = \min(v_0, a_c t), \quad (7)$$

where a_c is set to match the run-level mean measured acceleration during the mid-acceleration phase.

B2) Constant-deceleration (CD) baseline (stop-controlled approach): For Experiment 2, we consider a constant-deceleration stopping profile toward the stop line:

$$v_{CD}(d) = \sqrt{\max(0, v_{init}^2 - 2b_c(d_{init} - d))}, \quad (8)$$

where b_c is selected to match the run-level mean measured deceleration during the approach-to-stop phase and d denotes the longitudinal distance-to-stop-line along the planned path, obtained from the GNSS/INS trajectory projected onto the reference centerline. The initial distance d_{init} is taken at the run start time.

B3) Tracking baseline using a first-order lag: To separate behavioral mismatch from actuation dynamics, we also evaluate a purely dynamical baseline in which the measured speed is predicted using a first-order response model driven by

a reference trajectory. In addition to IDM, we test the same mapping when driven by the CA/CD baselines. This provides a simple check of whether the improvement achieved by the lag is specific to IDM or largely attributable to generic smoothing dynamics.

C. First-Order Response Mapping

To model the aggregate longitudinal response of the controller and actuators, we apply a first-order mapping between a reference speed trajectory $v_{ref}(t)$ and the realized speed $v(t)$:

$$\dot{v}(t) = \frac{1}{\tau}(v_{ref}(t) - v(t)), \quad (9)$$

where τ is an effective time constant. In the experiments, $v_{ref}(t)$ is taken as the IDM-predicted trajectory $v_{IDM}(t)$ or, for baseline comparisons, the CA/CD reference.

For numerical implementation at planner/controller step $\Delta t = 0.01$ s (100 Hz), Eq. 9 is discretized via forward Euler:

$$v_{k+1} = v_k + \frac{\Delta t}{\tau}(v_{ref,k} - v_k), \quad \tau > 0. \quad (10)$$

The initial condition is $v_0 = v_{meas}(t_0)$ at the trajectory alignment instant t_0 . Per-run τ is identified by minimizing RMSE

over bounded search $\tau \in [\tau_{\min}, \tau_{\max}]$ with $\tau_{\min} = 0.05$ s and $\tau_{\max} = 2.0$ s.

D. Statistical Inference and Generalization Protocol

To evaluate whether apparent performance gains are statistically meaningful and transferable, we use a paired-run analysis protocol:

- **Paired inference for error reductions:** for each run, we compute $\Delta\text{MAE} = \text{MAE}_{\text{raw}} - \text{MAE}_{\text{lag}}$ and $\Delta\text{RMSE} = \text{RMSE}_{\text{raw}} - \text{RMSE}_{\text{lag}}$. We report mean $\pm$ std, 95% confidence intervals, and paired significance tests.
- **Cross-trajectory τ validation:** we perform leave-one-trajectory-out evaluation, fitting τ on all runs except one and testing on the held-out run.
- **Fixed- τ vs per-run- τ :** we compare a global regime-level $\bar{\tau}$ against per-run fitted τ to quantify overfitting risk and transferability.
- **Effect size reporting:** we report normalized improvement ratios for MAE/RMSE to complement p-values.

This protocol directly addresses generalization concerns by distinguishing platform-level response structure from trajectory-specific calibration.

APPENDIX D RESULTS AND DISCUSSION

A. Parameter Sensitivity Analysis

To further investigate the robustness of the calibrated IDM parameters under varying driving regimes, a sensitivity analysis was conducted on both experimental settings. Figure 6 illustrates the MAE distribution across different parameter combinations, with the left panel corresponding to the free-flow experiment and the right panel to the stop-sign experiment.

In the free-flow experiment (Fig. 6a), MAE values are plotted for combinations of desired speed v_0 and maximum acceleration a . The results indicate a clear dependence of model accuracy on the acceleration parameter. Runs with lower acceleration values ($a = 0.5$ m/s²) exhibit substantially higher prediction errors ($\text{MAE} > 2$ m/s), whereas moderate acceleration levels ($a = 1.0$ - 1.5 m/s²) yield significantly improved fits to the empirical data. This trend suggests that the controller’s ability to reach target velocity smoothly without overshooting is best captured by intermediate acceleration gains, consistent with realistic longitudinal actuation dynamics. At higher accelerations ($a = 2.0$ m/s²), deviations begin to reappear, reflecting increased mechanical lag in the throttle response and slight discrepancies between theoretical and measured profiles.

The stop-sign experiment (Fig. 6b) explores the combined influence of time headway T and jam distance s_0 on IDM performance. Overall, the model achieves low error levels ($\text{MAE} < 1$ m/s) across most configurations, demonstrating stable predictive capability near the stop condition. However, the sensitivity map reveals a distinct basin of optimality around $(T, s_0) = (1.0, 1.5)$, where MAE reaches a minimum of approximately 0.53 m/s. Conversely, longer headways ($T >$

1.5 s) tend to increase the error, likely due to overestimation of deceleration onset relative to the actual stop position. These findings align with the physical interpretation of the IDM spacing function, where overly conservative time headways induce premature braking compared to the measured trajectory.

Across both experiments, the sensitivity results reinforce that the IDM captures longitudinal behavior most accurately when its parameters align with the vehicle’s natural response dynamics. The subsequent introduction of a first-order lag parameter τ in Eq. 9 further quantifies the control delay that bridges the theoretical and measured profiles. Together, these analyses provide a data-driven pathway for refining IDM-based controllers by explicitly linking perception and actuation uncertainties to parameter-level sensitivity.

B. First-Order Response Mapping Between Reference and Realized Trajectories

The comparisons above indicate that IDM, with reasonable parameter choices, replicates the main qualitative features of the measured trajectories. However, systematic discrepancies remain, especially in the timing and sharpness of accelerations and decelerations. Rather than attributing these mismatches solely to deficiencies in the behavioral model, we model them as arising from the combined dynamics of perception, planning, and control. This distinction is important when a behavioral model is deployed as a component of an autonomous agent: the model specifies the nominal behavioral response, whereas the realized response emerges from the interaction between that model and the system that supplies its state information and executes its commands.

In this work, we focus on a simple first-order transitional mapping that captures the dominant actuation and controller dynamics. Let $v_{\text{IDM}}(t)$ denote the idealized velocity trajectory produced by integrating the IDM acceleration law in Eqs. 1-2 under the same initial conditions as the experiments. The realized vehicle speed is modeled as the output of a first-order system driven by $v_{\text{IDM}}(t)$ via Eq. 9, where τ is an effective time constant representing the aggregate response of the longitudinal controller and powertrain.

Eqs. 1-2 together with Eq. 9 define a modified IDM-AV model in which the behavioral law (IDM) is filtered through the actuation dynamics characterized by τ .

Quantitative lag results for both experiments, including paired improvements and leave-one-trajectory-out transferability, are summarized in Table V; representative overlays are shown in Fig. 7. The main text (Sec. IV-C) reports the headline stop-sign reductions; here we retain the fitting details and supporting statistics.

Visual inspection of the trajectories further suggests the presence of perception and planning delays, manifested as a phase shift in the onset of braking. We do not introduce an explicit additional delay state in the mapping; instead, we quantify perception latency descriptively and isolate its impact via ablation.

Beyond actuation effects, perception also played a measurable role in shaping the observed trajectories. During the

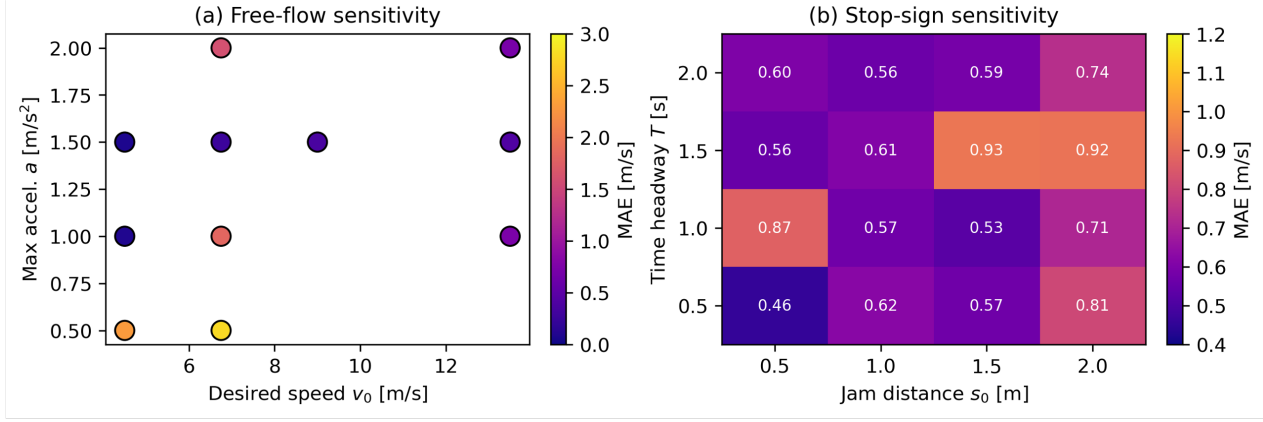


Fig. 6: Parameter sensitivity of IDM prediction error across both experiments.

stop-sign experiments, the vehicle relied on LiDAR-camera fusion and localization to estimate the distance to the stop line, which was modeled as a stationary virtual leader. Frame-level detection delays and minor range estimation noise introduced an average perception state latency of 0.33 ± 0.12 s before the planner received a stable leader state. This lag explains part of the phase shift observed between the IDM-predicted and measured braking onsets, confirming that perception timing can significantly affect model validation even when the underlying behavioral law is accurate.

For completeness, the same first-order lag formulation was also applied to the free-flow runs. Although perception delays play a smaller role in this regime, fitting τ captures the effective mechanical and controller dynamics during acceleration. Table V reports the fitted τ values, post-lag errors, paired improvements, and cross-trajectory validation results, and Figure 7 shows representative trajectories after applying the mapping.

These baseline comparisons serve two purposes. First, they contextualize the predictive accuracy of IDM relative to simple kinematic profiles that ignore state-dependent interaction structure and the closed-loop realization of perception-planning updates. Second, they help interpret the role of the first-order mapping: if the lag improves multiple references similarly, the dominant mismatch is likely due to actuation/controller dynamics; if the improvement is consistently larger for IDM, the IDM structure provides a more informative reference trajectory for the AV to realize.

The fitted values indicate a faster response in free-flow driving, where the system continuously accelerates toward the desired speed, and a slower effective time constant during braking, where the controller’s smoothing and perception-driven anticipation delay introduce additional lag. This difference confirms that the same first-order formulation can describe both acceleration and deceleration regimes with distinct but interpretable dynamic responses.

These results suggest that behavioral-model evaluation should distinguish between the nominal response prescribed by the behavioral law and the response ultimately realized by

the autonomous platform. The first-order mapping does not modify the IDM behavioral formulation; instead, it exposes how downstream system dynamics mediate the execution of the behavioral decision.

C. Simulation & Validation

To complement the closed-course trajectory analysis, two simulation-based diagnostics were conducted to determine whether the implemented planner preserves the qualitative traffic-flow behavior associated with IDM. Both diagnostics use the standard IDM formulation in Eqs. 1-2. Two variants are compared: Normal IDM, in which all vehicles follow the standard acceleration law, and Lag-aware IDM, in which the AV subpopulation follows the first-order response mapping in Eq. 9, while human-driven vehicles retain the standard IDM formulation.

Equilibrium behavior is first evaluated on a ring road by varying density while holding the number of vehicles fixed. Following a warm-up period, the steady-state mean speed $\bar{v}$ is measured and flow is calculated as $q = k\bar{v}$, where k is density. Multiple repetitions are conducted at each density to capture variability near capacity.

As shown in Fig. 8, both variants reproduce the expected free-flow, transition, and congested regimes. The lag-aware configuration largely preserves the free-flow branch but produces lower speeds and flows after breakdown, indicating that delayed vehicle response affects congestion recovery without changing the underlying IDM behavioral law.

Transient stability is evaluated by applying the same time-varying leader-speed disturbance to both model variants. The resulting shockwave diagrams in Fig. 9 illustrate the formation and propagation of stop-and-go waves through the platoon.

Both variants reproduce coherent backward-propagating congestion waves. However, Normal IDM produces sharper congestion bands and faster speed recovery, whereas the lag-aware configuration produces wider and more persistent low-speed regions. This behavior results from the first-order response mapping, which slows the realization of acceleration and deceleration commands and therefore reduces disturbance damping.

TABLE V: First-order lag fitting, statistical improvement, and cross-trajectory transferability.

Panel A: Fitted lag parameters and post-lag errors				
Experiment	τ [s]	# Runs	MAE [m/s]	RMSE [m/s]
Free-flow, well-matched	0.350 ± 0.120	7	0.481	0.723
Stop-sign, all runs	0.670 ± 0.190	16	0.511	0.632
Panel B: Paired improvement for stop-sign runs				
Metric	Mean improvement	95% confidence interval		Runs
Δ MAE	0.157	[0.09, 0.23]		16
Δ RMSE	0.235	[0.14, 0.33]		16
Panel C: Cross-trajectory transferability				
Configuration	MAE [m/s]		RMSE [m/s]	Evaluation
Per-run fitted τ	0.511		0.632	In-sample
LOTO fixed $\bar{\tau}$	0.548		0.689	Held-out

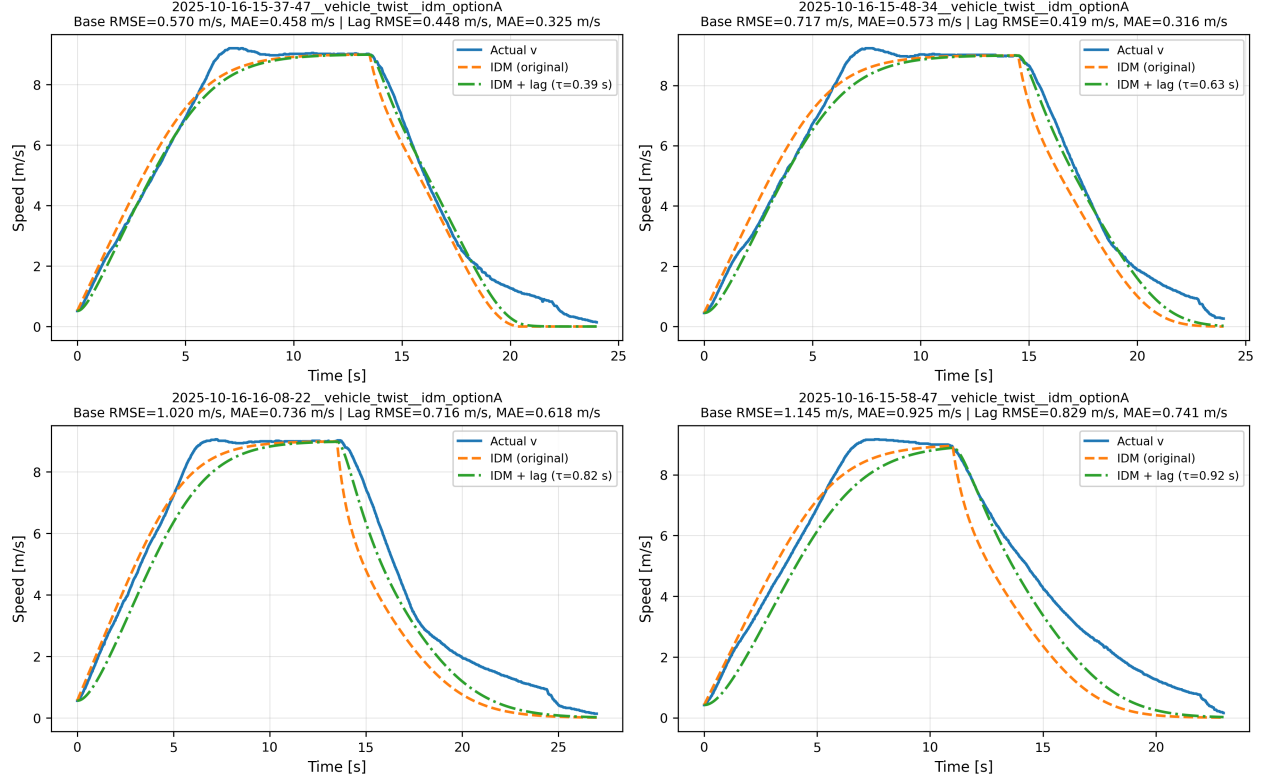


Fig. 7: Representative stop-sign approach showing measured speed, raw IDM prediction, and lagged IDM trajectory.

These findings are consistent with the fundamental diagrams, where the lag-aware configuration shows lower average speed and flow in congested conditions. Overall, simulation shows that the response mapping preserves the recognizable behavior of IDM while demonstrating that platform-level actuation dynamics can materially affect traffic stability and congestion recovery.

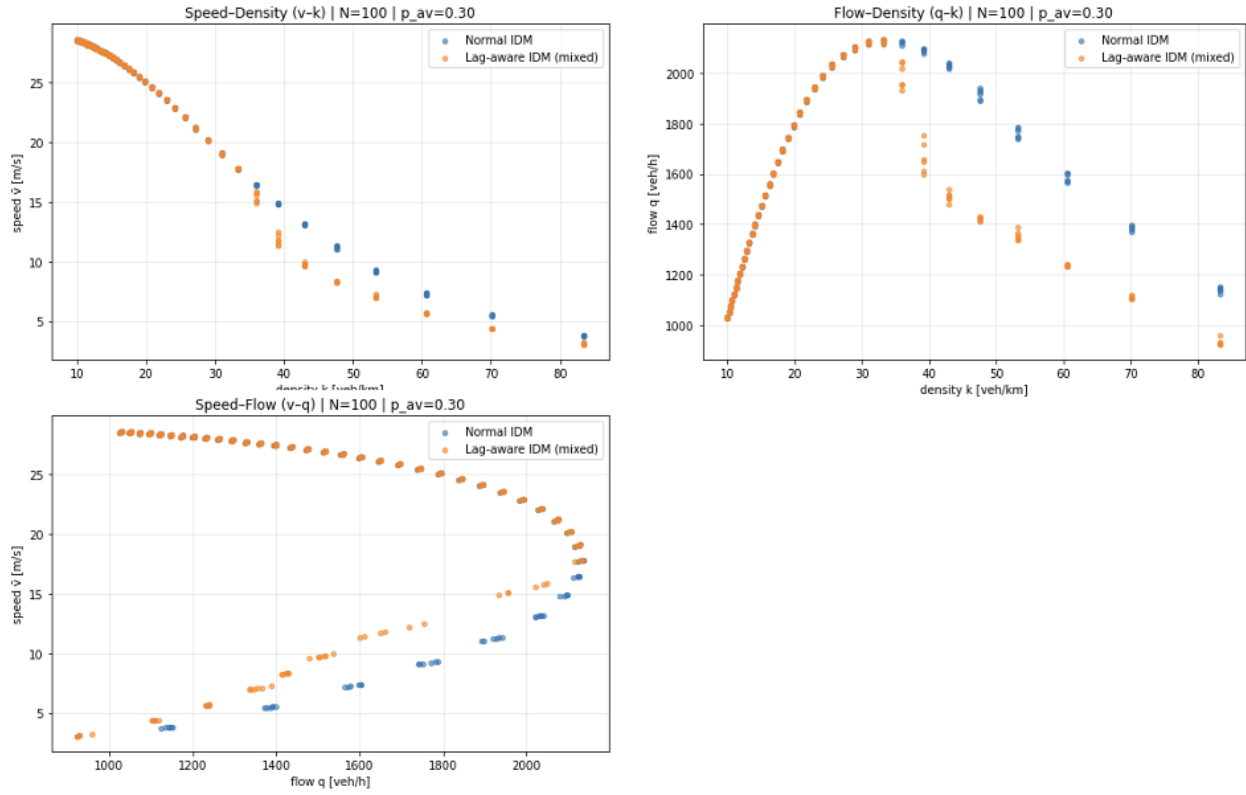


Fig. 8: Equilibrium fundamental diagrams from the ring-road experiment for Normal IDM and the lag-aware mixed-traffic configuration.

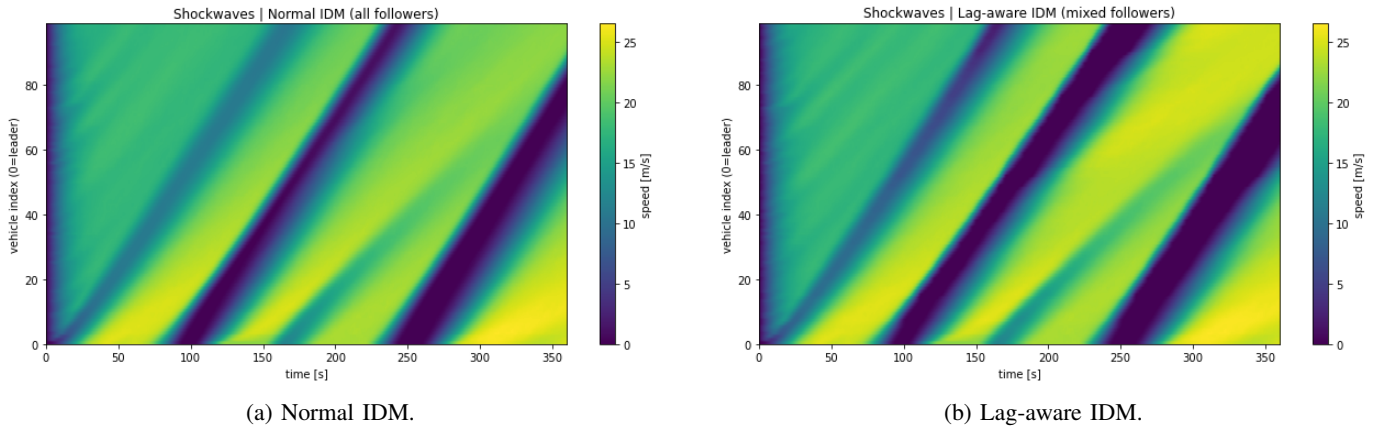


Fig. 9: Shockwave spacetime diagrams under the same forced leader disturbance.